

Problem Analysis and Design

Overview

In this deliverable, you will work in your group to **analyse your project idea** and produce a **design for your system** using an Object-Oriented Programming (OOP) approach. Your analysis must be based on the system workflow and storyboard that you presented in Deliverable 1. This deliverable consists of **two (2) tasks**:

1. a **Problem Analysis session** recorded as a video, and
2. a **Design Report** submitted as a PDF file.

Task 1: Problem Analysis (Video)

Conduct a **group discussion or brainstorming session** to analyse the problem of your project using an OOP approach. Base your analysis on the system workflow that you described in Deliverable 1.

What to Do During the Session

During the session, your group must:

1. **Identify the objects and classes** involved in your project.
 - List the main objects in your system (e.g., the things your system manages or works with).
 - From those objects, determine the classes and define their significant attributes and methods.
 - For each class, briefly state its purpose - what it represents and what it does in the system.
2. **Identify the class relationships.**
 - Determine which classes are involved in an association relationship (aggregation or composition).
 - Determine which classes are involved in an inheritance relationship.
 - **Justify each relationship:** explain why you chose that relationship type. For example: why is this an aggregation and not a composition? Why does this class inherit from another?

★ Guidance on identifying good classes:

A good class represents a meaningful concept or entity in your system. Ask these questions to help identify classes:

- What are the main things (objects) that this system manages or stores?
- What information does each object need to hold? (These become attributes.)
- What actions can each object perform or what can be done to it? (These become methods.)
- Are there two or more types of the same object with different behaviour? (These suggest inheritance.)

How to Conduct the Session

- **Collaborative, not a presentation.** The session is meant to show your group working through the analysis together. It should not look like a rehearsed presentation.

- **Conduct the session synchronously:** either online or face-to-face.
- **If online:** use an online meeting tool such as Webex, Google Meet, or Zoom. Turn your camera on and share your screen.
- **If face-to-face:** you may meet in person, but you must still video-record the full session.
- **Use an online whiteboarding tool** to write and organise your ideas during the session. Recommended tools: Whimsical, Google Jamboard, Miro, or any similar tool.
- **All members must actively participate:** every member should be seen contributing to the discussion (speaking, writing on the board, or responding to questions).
- **Upload the video to YouTube** and submit the URL to e-learning.

Task 2: Design Report (PDF)

Based on the findings from your analysis session, produce a **written design report**. The report must follow the structure below:

★ Report format and submission:

- The report must be submitted as a PDF file.
- Use clear headings for each section as listed below.
- Include your group name, member names, and section number on the first page.
- The report does not need to be long: aim for clarity and completeness over length.

a. UML Class Diagram

Provide a UML Class Diagram that includes:

- Key classes** with their significant attributes and methods.
- Relationships** between classes including association (aggregation or composition) and inheritance where appropriate.

Diagram tools:

You are encouraged to use a diagramming tool to create a clear, well-labelled class diagram. Recommended tools: draw.io (app.diagrams.net), Lucidchart, Visual Paradigm, or any similar UML tool.

Follow standard UML notation: class name in the top section, attributes in the middle, and methods at the bottom. Use the correct notation for visibility (+ public, - private, # protected) and for relationship types (solid line with open triangle for inheritance, diamond for composition/aggregation).

b. Class Descriptions

Briefly describe each main class in your diagram. For each class, include:

- Purpose:** what the class represents.
- Main responsibilities:** what the class does or manages.
- Role in the system:** how the class fits into and interacts with the overall system.

A short paragraph (3-5 sentences) per class is sufficient. Avoid simply repeating the attribute and method names, focus on explaining the purpose and role of the class.

c. Relationship Explanation

Explain the relationships between the classes shown in your UML diagram. For each relationship, provide a **justification**, explain why that relationship type is appropriate.

- i. **Inheritance:** identify the base class and derived classes and explain what is being shared or specialised.
- ii. **Association (Aggregation and/or Composition):** identify which classes are involved, state whether it is aggregation or composition, and justify your choice.

★ Justification tip:

Use this pattern when explaining a relationship: "[Class A] and [Class B] have a [relationship type] relationship because [reason]. If [Class A] is deleted, [Class B] [does / does not] still exist, which confirms it is [composition / aggregation]."

d. Design Rationale

Provide a short explanation of your overall design decisions. Discuss:

- i. Why certain classes were created.
- ii. Why specific relationships were chosen.
- iii. How your design improves modularity, maintainability, or reusability.

e. Design Assumptions

State any assumptions made during the design process. For example: simplified or idealised conditions used in your modelling, or any system features that were assumed but not yet confirmed.

If there are no significant assumptions, you may state "No major assumptions were made" and briefly describe the scope boundary of your design instead.

f. Design Limitations

Briefly describe any known limitations in your current design, or areas where the design could be improved in a future version.

Assessment

This deliverable carries **6%** weightage for the final grade of this course. The breakdown is as follows:

Criteria	Weightage
1. Problem Analysis (shown in the video): - Identify objects and classes (with attributes and methods) - Identify and justify class relationships <i>Active participation of all members during the session is also assessed.</i>	1% 1%
2. Design Report (submitted as PDF): - UML Class Diagram - Class Descriptions and Relationship Explanations - Design Rationale, Assumptions, and Limitations	1% 2%

Criteria	Weightage
<i>Marks are awarded for the quality of justification, not just completeness.</i>	
Teamwork Assessment 2 (TW2) - Peer and Self Assessment: Individual submission. This assessment is evaluated by your peers, yourself, and the lecturer. The lecturer's assessment is based on your visible involvement in the analysis video. Failure to submit will result in a ZERO mark .	1%
Total	6%

Teamwork Assessment 2 (TW2)

Each member must individually complete and submit **Teamwork Assessment 2 (TW2)**, which is a Peer and Self Assessment (PSA). This is an **individual submission**. It is not submitted as a group.

Important Notes for PSA:

- Download a copy of the Excel file.
- Edit on your own local copy.
- Submit your copy to e-learning.
- Use a separate copy for each PSA submission.
- Do not leave any empty fields. You must also rate yourself.
- Video guide on how to fill in the form:
https://drive.google.com/file/d/1E4xWmhkQo5ZMiGe3nR015i8Uub0wFeEN/view?usp=drive_link
- You may watch the video at 1.5× speed to save time (use the playback speed option in the video player settings).

Submission

- **Deadline:** Week 9, Friday, 15 May 2026, 23:59 MYT (UTC+8).
- **Late submission penalty:** 10% deduction for every 1 day late.

Group Submission

Only **one (1) member** per group needs to make the group submission.

- **Video (Problem Analysis):** Submit the YouTube video URL to e-learning.
- **Design Report:** Submit the report as a **PDF file** to e-learning. Name your file as follows:
GroupName_Section_D2.pdf (e.g., GroupAlpha_Sect04_D2.pdf).

Individual Submission (TW2)

- Each member must individually submit TW2 (Peer and Self Assessment) to e-learning.
- **Failure to submit will result in a ZERO mark** for TW2.